

Homeplate v1.0 — Verification Report (waves 1–2)

Date: 2026-08-12 · **Commit:** `01fe872` (wave 1–2 source is uncommitted working tree on top of it) **Scope of this build run:** waves 0–2 only (bootstrap, foundation, platform services, external adapters) per `docs/_generated/build-plan.md`. Waves 3–6 (marketplace features, trust & safety, React client) are intentionally **not built**; their requirements are reported below as *not implemented*, never as passes.

Status line

Command	Result
<code>npm test</code> (full suite, isolated — no concurrent lanes)	28/28 suites, 611/611 tests PASS (27.6 s). A second run with <code>--coverage</code> also 611/611.
<code>npx jest --coverage</code>	Statements 91.18 % (1935/2122) · Branches 81.12 % (1298/1600) · Functions 96.91 % (345/356) · Lines 92.12 % (1837/1994)
<code>npm run lint</code>	eslint clean; "All matched files use Prettier code style!"
<code>npm run build</code>	56 files parse cleanly; <code>createApp()</code> boots against <code>.env.example</code> ; "all checks passed"
Stub scan	<code>grep TODO/FIXME/not implemented</code> over <code>src/</code> , <code>scripts/</code> , <code>db/</code> → 0 hits ; no placeholder logic outside the ADR-007/011-mandated mock adapters, which production config refuses (<code>validateEnv</code> rejects <code>*_MODE=mock</code> under <code>NODE_ENV=production</code>).

The earlier multi-lane runs that showed 2–16 failures were caused by concurrent verifier sessions sharing one test database (see finding F-1); with a single runner the suite is deterministic and green — observed twice consecutively.

Verification honesty note: the suite is green *because* unbuilt requirements are asserted as absent (404-probe tests confirm the routes do not exist), not because those requirements pass. 20 of 35 requirement rows below are not fully verifiable yet.

Traceability (SRS Appendix B shape)

Status legend: **PASS** — acceptance criteria executed and met · **PARTIAL** — the wave-1/2 slice passes, the remainder is unbuilt · **NOT IMPL** — no implementing code (wave 3–6); where noted, the DB/adaptor *substrate* it will use is already verified · **UNTESTABLE** — cannot be verified by automation in this environment.

Req	Design element (SRS App. B)	Implementing file(s)	Test ID	Test status
FR-01 search	Search Service, Listing Service, Maps Adapter	— (maps substrate: <code>src/adapters/maps.js</code> , <code>src/lib/geoPrecision.js</code>)	TC-01	tested
FR-02 meal detail	Meal Detail View	—	TC-02	tested
FR-03 host page	Host Profile Service	— (<code>host_profiles</code> table + <code>PATCH /api/users/me</code> upsert exist)	TC-03	tested

Req	Design element (SRS App. B)	Implementing file(s)	Test ID	Test
FR-04 completion	Booking Service (completion)	— (schema: <code>db/migrations/0001_core_schema.sql</code>)	TC-04	test scenario
FR-05 reviews	Review Service	— (schema)	TC-05	test scenario
FR-06 messaging	Messaging Service	— (schema; outbox: <code>src/outbox/*</code>)	TC-06	test scenario
FR-07 safety alerts	Safety Alert Service, SendGrid Adapter	— (schema; <code>src/adapters/sendgrid.js</code> , templates registered)	TC-07, IT-04	test scenario acceptance acceptance
FR-08 moderation	Moderation Module	— (adapter: <code>src/adapters/llmModeration{,.mock}.js</code>)	TC-08, IT-03	test scenario mock
FR-09 eligibility	Eligibility Policy Service	<code>src/modules/eligibility/{policy,repo,middleware}.js</code>	TC-09	test scenario unit test
FR-10 registration	Registration & Email Verification	<code>src/modules/auth/*</code> , <code>src/outbox/handlers/emailVerification.js</code>	TC-10	test scenario registration test
FR-11 listing caps	Listing Service (MEHKO)	— (config: <code>src/config/</code> ; index: <code>db/migrations/0002</code>)	TC-11	test scenario capacity
FR-12 reservation	Booking Service (atomic capacity)	— (schema + config)	TC-12, LT-01	test scenario booking
FR-13 notifications	Transactional Outbox, Worker, Notification Adapters	<code>src/outbox/*</code> , <code>src/modules/notifications/*</code> , <code>src/adapters/{sendgrid,fcm,mockTransport}.js</code>	TC-13, RT-02	test scenario notification test outbox
FR-14 cancellation	Booking Service (capacity restore)	— (schema)	TC-14	test scenario booking

Req	Design element (SRS App. B)	Implementing file(s)	Test ID	Test Case
NFR-01 p95<500 ms @200 users	REST API, Redis Cache	—	LT-01	test volume test
NFR-02 10k users / 1k listings/day	Backend, PostgreSQL	scripts/seed.js (--volume), db/migrations/0002_indexes_constraints.sql	LT-02	test volume test
NFR-03 TLS 1.2+ only	Network Security Layer	src/server.js, src/middleware/security.js	ST-01	test security test
NFR-04 password hashing	Authentication Service	src/modules/auth/passwords.js	ST-02	test security test
NFR-05 login logout	Authentication (rate limiting)	src/modules/auth/rateLimit.js	ST-03	test security test
NFR-06 eligibility recompute	Eligibility Policy, Email Verification	src/modules/eligibility/*, src/modules/users/*	IT-02, TC-09	test verification test
NFR-07 WCAG 2.1 AA	Web UI (React)	— (no client/)	UT-01	—
NFR-08 logging/audit	Logging & Monitoring	src/lib/logger.js, audit calls in src/modules/auth, src/outbox/*	MT-01	test code test
NFR-09 degraded mode	External Adapters, Deferred-Work	src/adapters/*, src/lib/resilience.js, src/lib/cache.js, src/outbox/*	RT-01	test deployment test
NFR-10 moderation accuracy <5 % FP/FN	AI Moderation Service	—	IT-03	—
NFR-11 input validation	Input Validation Module	src/middleware/validate.js, src/schemas/*, src/lib/sanitize.js	ST-04	test security test
NFR-12 deletion/retention	Data Lifecycle Service	src/modules/media/service.js (erasure primitive), src/config (erasureDays=30)	ST-05	test security test

Req	Design element (SRS App. B)	Implementing file(s)	Test ID	Test
NFR-13 data minimization/encryption	Data Protection	<code>src/db/fieldCrypto.js, src/modules/users/repo.js</code>	ST-06	te se
AB-01 fake host/listing	Eligibility, Moderation, Reviews, Safety	—	TC-07..09, IT-03, ST-05	—
AB-02 booking hoarding	Booking Service, Eligibility, Logging	— (<code>config.booking.maxConcurrentPending=3</code>)	TC-12, ST-04, LT-01	—
AB-03 spam/scripted listings	Listing Service, Moderation, Validation, Auth rate limit	—	IT-03, ST-04, ST-03, TC-11	—
AB-04 abusive content	Moderation, Safety, Logging	—	IT-03, TC-05/06/08	—
AB-05 account takeover	Auth (hashing, rate limit), TLS	<code>src/modules/auth/{sessions,rateLimit,passwords}.js</code>	ST-01/02/03	te se
AB-06 injection	Input Validation Module	<code>src/middleware/validate.js, src/lib/sanitize.js</code>	ST-04	te se
AB-07 duplicate accounts / MEHKO evasion	Registration, Listing Service, Logging	<code>src/modules/auth/*</code>	TC-10, TC-11, MT-01	te re
AB-08 scraping personal data	Sessions, Eligibility, Data minimization	<code>src/modules/auth/sessions.js, src/adapters/maps.js</code> (coarse cache)	ST-06, ST-04	te se te ma

ADR conformance (primary signal for this stage): ADR-001/003 (outbox atomicity, no inline adapters, IDs-only payloads) **PASS**; ADR-005 (maps cache/degraded) **PASS**; ADR-006 (single auth + single eligibility policy) **PASS**; ADR-007 (provider-agnostic, mock-in-suite, zero provider/key literals) **PASS**; ADR-011 (push default-false, attempt rows, no live sends) **PASS**; Redis role audit (only `hp:session|ratelimit|cache:` namespaces) **PASS**. ADR-002/008 not implementable yet (wave 4); ADR-004/009/010 **partial** — mechanisms proven, their wave-3/4 callers absent.

Open findings

Resolved since the last round (verified on disk and by the green suite this session): **ITA-02 / RTLT2-F1 / COV-05** — `GET /health` now carries `validate({ query: noInput })` (`src/app.js:114`), NFR-11 enumeration passes; **RTLT2-F2** — the bootstrap probe key was renamed to `hp:cache:bootstrap-probe` and is deleted in `finally` (`tests/unit/bootstrap.test.js:85-91`), adr-conformance namespace scan passes.

F-1 (minor, test infrastructure) — Concurrent jest invocations share one test database and corrupt each other

Consolidates TCB-02, MTQ-04, COV-06, ADRCONF-F02. **Requirement:** none (verification reliability; NFR-02 signal quality). **Reproduction:** run two jest processes against the default `homeplate_test` DB / Redis DB 1 concurrently — each `globalSetup` drops/reseeds the schema under the other's live queries. Observed repeatedly during multi-lane verification: wandering failure sets (13–16 failures that change between runs), a `deadlock detected` in `DROP SCHEMA`, a >600 s `globalSetup` hang, and a foreign worker delivering another test's outbox job. **Current state:** `tests/helpers/globalSetup.js` still has no cross-process guard (no advisory lock). The `TEST_DATABASE_URL` / `TEST_REDIS_URL` / `OBJECT_STORAGE_BUCKET` overrides are honoured and proven to give clean isolation. **Proposed fix:** take `pg_advisory_lock(hashtext('homeplate_test_suite'))` in `globalSetup` (release in `globalTeardown`) so runs `serialize`, or derive per-run DB/Redis/bucket names; at minimum document that concurrent lanes **MUST** set the overrides. **Why still open:** does not affect a single-runner CI (two consecutive solo runs green, 611/611); fix is test-infra hardening the team should land before the next multi-lane verification round.

F-2 (minor, test) — `bootstrap.test.js` hardcodes the Redis DB index, defeating the documented override

COV-07. **Requirement:** none. **Reproduction:** run the suite with `TEST_REDIS_URL=redis://localhost:6379/5` → `tests/unit/bootstrap.test.js:73` fails: `expect(process.env.REDIS_URL).toMatch(/\/1$/)`. **Proposed fix:** assert isolation, not the literal default — parse the URL and require a non-dev index (`!== 0`) instead of exactly `/1`. **Why still open:** invisible on the default environment (passes there); only bites the F-1 workaround. Fix alongside F-1.

Not verifiable in this environment — manual actions for the team

1. **NFR-01 / LT-01 real load test** — the specified 200-VU, ≥5-min k6 scenario over search/listing/host endpoints cannot run: the endpoints do not exist (wave 3). Also note **k6 is a host prerequisite** (like docker) — it is not repo-installed; the smoke run used the official grafana/k6 v2.2.0 binary. After wave 3: run `npm run test:load` scaled to the LT-01 scenario on production-like hardware and record the p95.
2. **NFR-02 latency-at-volume** — DB half is done (seed + indexes + EXPLAIN); re-run LT-01 against the volume seed once wave-3 endpoints exist.
3. **UT-01 5-participant moderated usability study** — human activity, both roles (guest booking flow, host listing flow); schedule after the wave-5/6 client ships and triage findings before CDR (SRS §4.5).
4. **NFR-07 axe-core audit + keyboard traversal** — requires the React client (waves 5–6).
5. **NFR-10 / IT-03 measurement run** — requires the wave-4 pipeline, the ≥200-item synthetic eval set under `tests/fixtures/moderation-eval/vN/`, a **live** Gemini call (the only permitted live call, model id recorded), and a **recorded human label sign-off** (reviewer, date, set version) per ADR-008. Until then NFR-10 is unclaimable by policy.
6. **Live TLS scan of the deployed endpoint** — the dev listener was scanned in-process (TLS 1.2 negotiated, 1.1 refused); an external scan (e.g. `testssl.sh`) of the real deployment is a deploy-time manual check.
7. **OWASP ZAP baseline scan** (AB-06 acceptance) — run against a deployed instance once wave-3/4 input surfaces exist.

8. **ADR-007 open action** — record a finding on the Gemini free-tier data-use terms *before* any real user content is sent to the provider (wave-4 gate; the pipeline will scan private messages).

Measured numbers

- **NFR-01 p95:** *not measured* — *no pass claimed* (endpoints unbuilt). Recorded proxies, explicitly not NFR-01 evidence: `GET /api/users/me` (Redis session + PG read) at 200 concurrent clients for 10 s → ~6,100–8,900 req/s, p50 ≈ 31.5 ms, **p95 33.8–42.7 ms**, p99 46.6–77.5 ms, errors 0. k6 smoke `GET /health`: 150/150 checks, p95 3.83 ms, 0 % failed.
- **NFR-02 volume:** 10,000 users / 1,000 approved listings on one America/Los_Angeles day / 1,000 bookings seeded; EXPLAIN ANALYZE on FR-01 search predicates: day+cuisine 0.019 ms, geo-box 0.022 ms, zero sequential scans on `listings`; required indexes on `(scheduled_start)`, `(host_id, local_date)`, `(moderation_status)`, `(cuisine)`, `(coarse_lat, coarse_lng)` all present.
- **NFR-05 lockout:** attempts 1–5 with wrong password → 401 each; **attempt 6 with correct credentials** → **429** with numeric `Retry-After`; counter TTL ≤ 600 s in Redis (account+IP keys); window elapse restores login; success resets the account counter.
- **NFR-04 hash parameters:** `$argon2id$v=19$m=19456,t=2,p=1$...`; fallback bcrypt cost ≥ 12 (executed under forced module failure).
- **NFR-10 FP/FN: not measured; labeled-set size 0** (no eval set exists). No provisional numbers reported anywhere (verified: no results file exists).
- **NFR-12 erasure window:** `config.privacy.erasureDays = 30` (config only — no lifecycle job yet, so end-to-end erasure-window coverage is **0 %**). Media erasure primitive measured: 2 owned objects → exactly 2 `deleteByKey` calls → 0 `media_objects` rows remain, subsequent fetch 404s.
- **Coverage (full suite):** statements 91.18 %, branches 81.12 %, functions 96.91 %, lines 92.12 % over `src/** + scripts/**`. Remaining uncovered functions are child-process entry points (server/seed/migrate CLI mains) behaviorally exercised via spawned processes, and the live SendGrid/FCM provider branches that ADR-011 forbids the suite from reaching.

Bottom line for CDR

Waves 1–2 are **done and proven**: config/observability/validation foundation, TLS-only HTTP layer, auth + sessions + rate limiting, registration/email verification, the eligibility policy, the transactional outbox + worker with exactly-once recovery, all five external adapters with degraded modes, field encryption, and the full DB schema whose integrity constraints already make overbooking, single-party completion, double-restore, and same-day duplicate listings impossible at the data layer. Every gate (`npm test`, lint, build) is green.

Not done, by design: FR-01..08, FR-11/12/14 flows, all four AB-01..04 misuse defenses, moderation, privacy lifecycle, and the entire React client. NFR-01, NFR-07, NFR-10 have **zero** claimable evidence. Anyone reading a green CI badge as "requirements met" would be wrong — this table is the authority.